Element J

In the course of refining and validating the design of the Komodo dragon containment system, multiple stakeholders and field experts were consulted to provide feedback and evaluate the system's functionality, safety, and practicality. This external evaluation was crucial for identifying potential design flaws, confirming the system's effectiveness, and ensuring that the design met the required performance and safety standards. The stakeholders consulted included zookeepers, experts in mechanical engineering, and experienced designers with backgrounds in animal containment systems.

Zookeeper Evaluation

One of the most important aspects of the design evaluation came from the zookeepers who would be the end-users of the containment system. Their feedback focused on the usability of the system, particularly the guillotine door, the box connection system, and then the side panels. The zookeepers raised concerns about the ergonomics of the guillotine door, noting that its sliding mechanism, while functional, was somewhat rough in practice, especially when operated multiple times a day. They suggested that the door should move more smoothly, without requiring excessive force to slide into place, but that it was a minor issue compared to the other issues present. We

Additionally, the zookeepers highlighted the need for the system to be easily adjustable. They expressed a preference for a design that allows for quick adjustments of the box connection system, especially when working with different sizes of animals or when needing to modify the enclosure layout on short notice. Their feedback emphasized the importance of ease of use and time efficiency, especially during emergency situations.

The final piece of feedback from the zookeepers focused on safety. They emphasized the importance of ensuring that all mechanisms, including the guillotine door and box connections, were secure enough to prevent any accidental openings or failures while interacting with the Komodo dragons. Their concern was that the system should be fail-safe, ensuring that the animals would not escape under any circumstances. To be honest, their biggest concern was that we wouldn't have a finished product by the time but for our specific

Mechanical Engineering Expert Evaluation

The system was also evaluated by a mechanical engineering expert, who assessed the structural integrity and the feasibility of the system's components. The expert's feedback was particularly focused on the strength and durability of the box connection system and the toggle clamps used to secure the boxes together. We worked together for several weeks brainstorming different ideas that could work for this design with the limited funding, but this was the best idea we could come up with. We ran some calculations on the forces exerted on the system during typical use, determining that the system needed to be reinforced with stronger materials to ensure it would hold up under extended use.

Additionally, the engineering expert provided insight into the guillotine door's mechanism. The expert suggested that the current sliding mechanism might be improved by incorporating a track system that would help guide the door more smoothly, thereby reducing the friction encountered during operation. They also recommended testing the door's durability over extended periods to assess how the materials and moving parts would wear over time, especially in high-use scenarios.

Synthesis of Evaluations

The feedback from these diverse evaluators was synthesized to address specific concerns related to design requirements. Several common themes emerged from the evaluations:

1. **Usability and Ease of Operation**: Both the zookeepers and the mechanical engineer emphasized the importance of ease of use, particularly for the guillotine door. The door's sliding mechanism required further refinement to reduce friction and make it smoother to operate. Modifications,

such as adding a track system, were suggested to address this concern. Additionally, the box connection system needed to be more user-friendly, with easier adjustments and more secure locking mechanisms to ensure that the boxes would not accidentally come apart.

2. **Safety**: Safety was a priority across all evaluations. The zookeepers emphasized that the system must be foolproof, ensuring that the Komodo dragons could not escape under any circumstances. Both the designer and mechanical expert agreed that the toggle clamps and box connections needed to be more robust and that materials should be carefully selected to prevent failures over time.
3. **Durability and Strength**: The mechanical engineer's review highlighted the need for stronger materials in the box connection system to ensure the system's long-term reliability. The designer also noted the importance of considering corrosion-resistant materials to avoid degradation from environmental factors, such as humidity.
4. **Design Flexibility and Modularity**: The designer's evaluation brought attention to the potential for future modifications and scalability. While the current design was functional, considering future needs and the ability to modify the system for different animals or habitats was an important recommendation.

The evaluations provided a thorough and specific analysis of the design, with actionable suggestions for improvement. These evaluations led to a better understanding of how the design might perform in real-world conditions and have guided decisions regarding necessary modifications to the prototype.

The external evaluations provided by qualified stakeholders and field experts have significantly contributed to refining the design of the Komodo dragon containment system. The feedback was consistently detailed and specific, focusing on key areas such as usability, safety, durability, and adaptability. As a result, the design will be adjusted to improve the sliding mechanism, increase the strength of the box connections, and incorporate more durable, corrosion-resistant materials. Additionally, the evaluations highlighted the need for a more user-friendly, modular system that can be adapted over time. Moving forward, these insights will be used to finalize the design and ensure that it meets all of the specified requirements for the safe and effective containment of Komodo dragons.